Achievements Screen Spec

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Overview

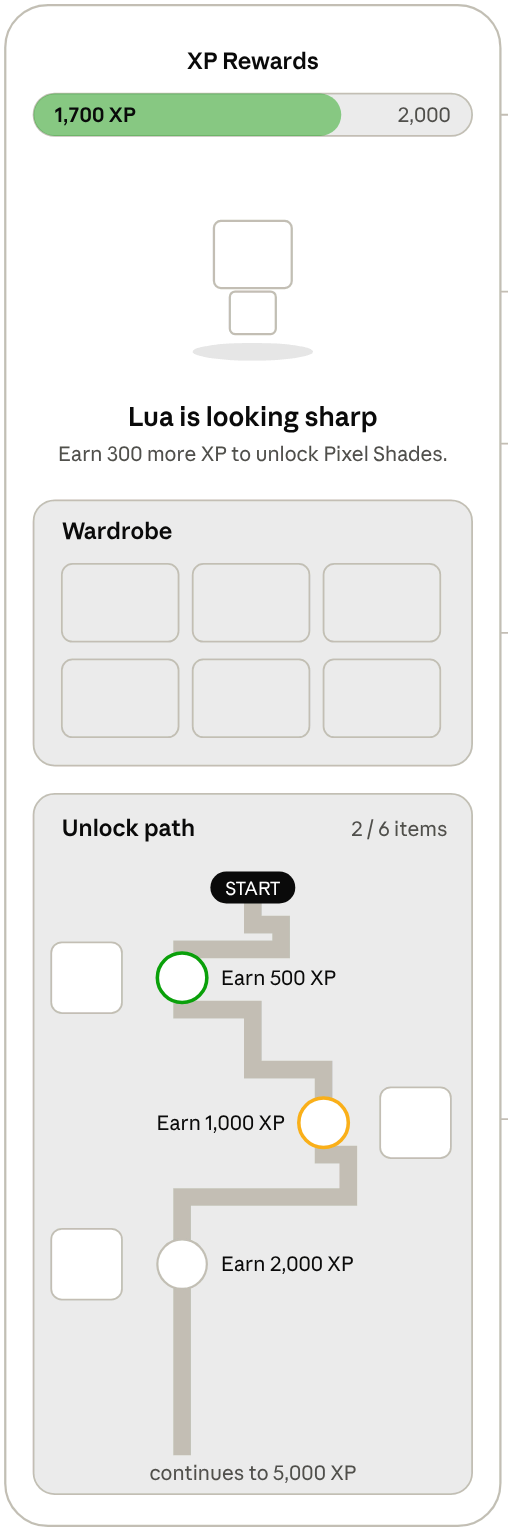
The achievements screen turns XP into wearable items for the player's pixel avatar: one item unlocks at each milestone from 500 to 5,000 XP. Players see their avatar, their XP and the next item, a wardrobe of items they own, and a staircase path of milestones with an animated preview of each item.

The reference build is the **Lua XP Rewards** prototype (open it). It is private until Lefika shares it. Where this spec and the prototype disagree, ask before building.

Screen layout

The screen is one scrolling column on every device, in the app's light style: title and XP bar, avatar, status line, wardrobe card, then the unlock path card. On larger screens it stays phone width (about 430 px), centred, like the app's other screens.

On every device the screen is one phone-width column



XP bar
Total inside, next milestone at the right

Avatar
On white; idle motion, toast, tap to hop

Status line
Mood headline, XP to the next item

Wardrobe
6 items in 3 columns
Tap to wear or take off

Unlock path
Staircase pixel road, START to 5,000 XP
Badge on the road, label on the inner side
Animated preview on the outer side

achievements screen on a phone · 4 regions, top to bottom

The background is white with 16 px side margins. The wardrobe and the path sit in light grey cards with 18 px corners, like the stats card on the Health Bar screen.

Avatar stage

The avatar sits large on the white background with no card behind it, the way Lua appears on the Health Bar screen, wearing everything currently equipped. The stage area is full column width and slightly shorter than it is wide (height about 0.92 of the width).

- **Avatar:** the 12×16 pixel sprite, centred, feet on a soft shadow ellipse about 88% of the way down. Always scale with nearest-neighbour filtering, never smoothed.
- **Idle loop:** the head dips gently (about a quarter of an art pixel, 2.6 s cycle), the hands sway the opposite way, and the eyes blink for 0.13 s every 2.2 to 5.4 s. Every few seconds the eyes glide one pixel left or right and back.
- **Tap:** a small hop with happy eyes, about 0.85 s: squash down, rise, land with a squash, settle. Ignore taps while a hop is playing.
- **Toast:** a dark pill (#1d1b24, white text) at the top of the stage with the item icon and text such as "Unlocked Pixel Shades!". It slides in and hides after 2.6 s.
- **Sneakers:** the soles add 2 art pixels under the feet, so the whole avatar rises 2 art pixels while sneakers are worn (eased, not a jump).

XP bar and status line

The XP bar sits at the top under the small centred title "XP Rewards", styled like the health bar. A status line under the avatar says how Lua is doing and what comes next.

- **XP bar:** a 30 px tall, fully rounded pill with a light grey track (#e6e6ea). It fills with a green-to-yellow gradient (#22a24a to #e8c21b) for progress from the previous milestone to the next, so at 1,700 XP it is 70% full (1,000 to 2,000).
- **Inside the bar:** the total on the left ("1,700 XP", thousands separated) and the next milestone on the right in grey ("2,000"). When XP is earned, the total counts up over 1.1 s, easing out.
- **Status headline:** large and bold, centred: "Lua is just getting started" with no items, "Lua is looking sharp" with some, "Lua has every item!" with all six.
- **Status line:** grey, centred: "Earn 300 more XP to unlock **Pixel Shades**." After the last item: "Every reward on the path is unlocked."

Wardrobe

The wardrobe is a 3-column grid of all six items, in a light grey card (#f3f3f5) headed "Wardrobe" with the hint "Tap an item to wear it" on the right; tapping an owned item puts it on or takes it off. Each item is a white tile (1.5 px border #e3e2e9, 12 px corners) with the item icon, its name and a one-line status.

State	Card look	Status text	On tap
Locked	Icon grey and darkened	Locked · 2,000 XP	Toast: "Earn 2,000 XP to unlock"
Owned	Normal	Tap to wear	Puts the item on
Wearing	Green border (#22a24a), pale green tile (#eef8f1)	Wearing (green)	Takes the item off

- **One item per slot.** Putting on an item replaces whatever is in its slot. The beanie and the crown share the head slot.
- **Put on:** the item pops onto the avatar, scaling from 1.35 to 1 over 0.35 s.
- **Take off:** the item floats up about 1.6 art pixels and fades out over 0.35 s.

Unlock path

The path sits in its own light grey card below the wardrobe. It is a pixel road that runs from a dark START pill at the top down through the six milestones in a staircase: straight vertical and horizontal runs with square corners, never curves or diagonals. The card is headed "Unlock path", with a count such as "2 / 6 items" on the right.

Road shape

- Build it on a tile grid: 80 columns across the path area. Milestones alternate between columns 30 and 50, 28 rows apart; START is at column 40.
- Each stretch to the next milestone goes down 8 rows, sideways to a detour column, down 10 rows, sideways to the next milestone's column, then down to it. Detour columns in order: 48, 40, 54, 38, 44, 26.
- The road is 6 tiles wide with a 1-tile dark border (#4d485c, the sprite outline colour) all round, square at the corners. Unearned road is a light grey checker (#e9e7ef / #dedbe7) in 2×2-tile squares.

Progress fill

- Tiles turn green in route order as XP rises. Between two milestones the fill is proportional: 1,700 XP fills 70% of the 1,000 to 2,000 stretch.
- Filled road is a green checker (#22a24a / #30b257). The 2 tiles at the front edge are light green (#9be8b3).
- Each newly filled tile flashes white (#ffffff) for 140 ms, then bright green until 420 ms, then settles. Tiles already earned when the screen opens do not flash.

Milestones

- **Badge:** a 56 px white circle on the road at each milestone, showing the item icon. Done = green ring and a green check. Next = gold ring (#e2b324) that pulses every 1.8 s. Locked = pale grey circle (#ecebf1), faded grey icon, small lock.
- **Label:** on the side of the badge facing the middle of the screen: "Earn 1,000 XP" in the pixel font, the item name below it (green once unlocked), on a backing in the card colour so it stays readable over the road.
- **Preview:** on the outer side of each milestone, alternating left and right, a white rounded tile (1.5 px border #e3e2e9, 14 px corners) playing the avatar wearing only that item. A tag under it reads "Preview", or "Unlocked" on green once earned.

Item	Preview pose
Sneakers	Quick happy hops with a squash on each landing
Denim Pants	Side-to-side strut, eyes glancing the way it leans
Pixel Shades	Cool nod, then a white shine slides across the lenses
Cozy Beanie	Shivers in falling snow, then a happy, cosy face
Striped Scarf	Eyes closed, leaning into gusts of wind, scarf flapping
Gold Crown	Proud bounce with gold and white twinkles round the crown

Unlock flow and rules

An item unlocks the moment the counting-up XP number passes its threshold, and it is put on automatically.

XP	Item	Slot
500	Sneakers	Feet
1,000	Denim Pants	Legs
2,000	Pixel Shades	Face
3,000	Cozy Beanie	Head
4,000	Striped Scarf	Neck
5,000	Gold Crown	Head (replaces the beanie)

What happens when XP is earned:

1. The XP number counts up over 1.1 s, and the road fills tile by tile in step with it.
2. When the count passes a threshold, the item unlocks:
    a. The avatar hops with happy eyes.
    b. The item drops onto the avatar from about 9 art pixels above with a bounce (0.9 s). Anything already in that slot fades out.
    c. Pixel sparkles (white, gold, light blue, green) burst from the item as it lands.
    d. The toast shows "Unlocked <item>!".
    e. The badge turns done, the next badge gets the gold ring, the wardrobe card shows Wearing, and the preview tag switches to Unlocked.
3. If one gain crosses several thresholds, the unlocks play in order, 1 s apart.

Visual style

The screen uses the app's light theme, matching the Health Bar screen: white background, light grey rounded cards, green for progress and the main button, and one pixel font throughout.

Use	Colour
Screen background, item tiles, previews, badges	#ffffff
Cards (wardrobe, unlock path)	#f3f3f5, 18 px radius, faint shadow
Tile and card borders	#e3e2e9
Main text, START pill, toast	#1d1b24
Secondary text	#6c6879
Green (progress, wearing, unlocked, main button)	#22a24a, pressed #1b8a3e
XP bar gradient	#22a24a to #e8c21b on a #e6e6ea track
Next-milestone gold	#e2b324

- **Sprite colours:** outline #4d485c, white #eeebf8, lavender #cfcbdd, purple #ac9dc7. The item colours are baked into the sprite sheets.
- **Effects on white:** sparkles use purple, gold, blue and green; snow is a soft blue-grey (#a9b8dc); crown twinkles are gold and orange. Plain white effects would vanish on the white background.

Assets

All art comes as ten 2048×2048 sprite sheets in `lua_sprite_sheets.zip`, each a 4×4 grid of 512 px cells with one animation per row. Every cell is a 32×32 grid of art pixels at 16 px each, and the avatar's feet sit on y = 464 in every cell of every sheet. The zip also has `luaSprites.ts` (requires plus row and cell maps) and `sprites.json`.

Sheet	Rows
`avatar_body.png`	idle, blink & glance, hop, happy
`item_sneakers.png`, `item_jeans.png`, `item_shades.png`, `item_beanie.png`, `item_scarf.png`, `item_crown.png`	Same rows and frames as the body: stackable layers
`preview_poses_1.png`	Previews: sneakers, jeans, shades, beanie
`preview_poses_2.png`	Previews: scarf, crown; unlock sparkle burst; empty

Sheet	Rows
`item_icons.png`	Icons: unlocked in rows 0 to 1, locked in rows 2 to 3

- **Dressing the avatar:** stack the body sheet and the worn item sheets inside one animated view, bottom to top: body, jeans, sneakers, scarf, shades, then beanie or crown. Each item sheet holds only the pixels that item changes, so the layers line up with no offsets.
- **Sneakers lift:** while sneakers are worn, raise the whole stack by 32 cell px (scaled to the on-screen size).
- **Road:** not in the sheets. Draw it in code from the tile grid in the Unlock path section.

React Native implementation notes

Run every animation on the UI thread with Reanimated, the same way `HealthStatusSprite` steps frames with a transform; never step frames with React state.

- **Frame stepping:** reuse the `HealthStatusSprite` pattern (`withRepeat` + `withTiming` to 4, `Math.floor` , `translateX` / `translateY` on the sheet). Play the 4-frame rows at about 6 to 8 fps.
- **Smooth motion on top:** the sheets only have 4 frames per row. Add the bob, squash and stretch, item drop bounce, pop and fade with Reanimated transforms on the stacked view.
- **Crisp pixels:** draw the sheets with nearest-neighbour filtering. React Native Skia (`FilterMode.Nearest`) does this. Plain `Image` will blur unless the art is shown at an exact multiple of its size.
- **Memory:** each sheet uses about 16 MB once decoded, so a fully dressed avatar (7 layers) is around 110 MB. If that is too much, ask for quarter-size sheets (512×512) and scale them up with Skia.
- **Road:** compute the tile list once from the route, then draw it with Skia rects. Only redraw while the fill is moving or tiles are flashing.
- **Reduce Motion:** when `AccessibilityInfo.isReduceMotionEnabled()` is true, turn off the idle bob, hops, bounces, sparkles, tile flashes and the badge pulse.
- **Accessibility:** give the avatar, each wardrobe card and each milestone a label, for example "Pixel Shades, locked, unlocks at 2,000 XP".

Prototype-only parts and open questions

The four demo buttons in the prototype (+150 XP, +500 XP, Max out, Reset) only simulate XP and are not part of the app, and neither is its starting state of 1,200 XP.

- [ ] Where does XP come from, and when should the screen hear about new XP (live, or on next open)?
- [ ] Where are unlocked and equipped items stored, and do they sync across devices?
- [ ] What happens past 5,000 XP: more milestones later, or the path ends?
- [ ] Should unlocking always put the item on, or ask first?
- [ ] Should tapping a preview or badge do anything, such as open the item?
- [ ] Does the avatar also appear on other screens wearing the same outfit?